

Invasive Alien Species of India: GRIIS Checklist Cleaning and Analysis

Portfolio project by **Soham Sharma**.

An end-to-end, reproducible workflow: take the raw GRIIS checklist for India (Darwin Core Archive tables), clean and merge it with one script, and summarise the invasive species by kingdom, family and habitat.

Key findings

Metric	Value
Records in the India checklist	2,176
Flagged <code>Invasive</code> (any taxonomic status)	282
Flagged <code>Invasive</code> and <code>ACCEPTED</code> (analysis set)	253
Not evaluated for invasiveness	1,894

Workflow

Step	File
1. Source data (Darwin Core Archive tables)	<code>raw_taxon.csv</code> , <code>raw_speciesprofile.csv</code> , <code>raw_distribution.csv</code>
2. Processing script	<code>clean_griis_data.py</code>
3. Final clean data	<code>griis_india_clean.csv</code> (all 2,176 records), <code>griis_india_accepted_invasive.csv</code> (253 records)
4. Charts	<code>chart_family.png</code> , <code>chart_habitat.png</code> , <code>chart_kingdom.png</code>

What the script does

1. Checks that `id` is unique in each table.
2. Merges the Taxon core with the SpeciesProfile and Distribution extensions on `id` (1:1, validated).
3. Trims whitespace and recodes empty `isInvasive` values as `Not evaluated`.

4. Keeps records with `isInvasive = Invasive` and `taxonomicStatus = ACCEPTED`.
5. Writes the clean CSVs and the charts.

Reproduce

```
git clone <your-repo-url>
cd <your-repo-name>
pip install -r requirements.txt
python clean_griis_data.py
```

Running it overwrites the clean CSVs and charts with identical results.

Results

Kingdom

 Kingdom

Habitat

 Habitat

Families

 Families

Ecological interpretation

- **Kingdom.** Of the 253 accepted invasive records, 237 (93.7%) are plants (Plantae) and 16 (6.3%) are animals (Animalia). The animals are mostly fishes (14 ray-finned fishes), plus one insect (an ant family, Formicidae) and one mammal (a deer family, Cervidae).
- **Habitat.** 229 records (90.5%) are terrestrial. The remaining 24 are freshwater or combine freshwater with brackish, marine or terrestrial habitats. Most of the freshwater-linked records are fishes.
- **Families.** The list spans 73 families. Asteraceae (44 records), Fabaceae (33) and Solanaceae (19) lead, followed by Amaranthaceae and Poaceae (14 each). Together the top five families account for about 49% of the analysis set.
- **Orders.** Asterales (44), Fabales (33), Solanales (29), Caryophyllales (23) and Lamiales (20) are the largest orders.
- **Overall pattern.** The invasive alien record set is dominated by terrestrial plants from a few large families, with a smaller freshwater fish component.

Limitations

- **"Not evaluated" is not "non-invasive."** GRIIS leaves `isInvasive` empty for 1,894 records because they have not been assessed.

- **Counts are records, not always unique species.** *Ipomoea aquatica*, *Ipomoea cairica* and *Kalanchoe pinnata* each appear twice in the checklist, once flagged Invasive and once not evaluated. Only the Invasive row is in the analysis set.
- **This is a descriptive summary of a checklist.** It counts records; it does not measure abundance, spread or ecological impact.
- **Habitat is a coarse field** and can list several habitats separated by | .

Data source and attribution

Sankaran K V, Khuroo A A, Raghavan R, Molur S, Kumar B, Wong L J, Pagad S (2021). *Global Register of Introduced and Invasive Species - India*. Version 1.5. Invasive Species Specialist Group ISSG. Checklist dataset. <https://doi.org/10.15468/uvnf8m> (via GBIF.org)

Background on the GRIS method: Pagad S, Genovesi P, Carnevali L, Schigel D, McGeoch M A (2018). Introducing the Global Register of Introduced and Invasive Species. *Scientific Data* 5: 170202. <https://doi.org/10.1038/sdata.2017.202>

The raw_*.csv files were extracted from a combined workbook derived from this dataset. The data is used under **CC BY 4.0**

(<https://creativecommons.org/licenses/by/4.0/>), which requires attribution to the original authors as given above. Please confirm the licence and version on the GBIF dataset page before redistributing.

License

Code (`clean_griis_data.py`): MIT, see `LICENSE.txt` . Copyright (c) 2026 Soham Sharma.
Data: CC BY 4.0, attribution as above.